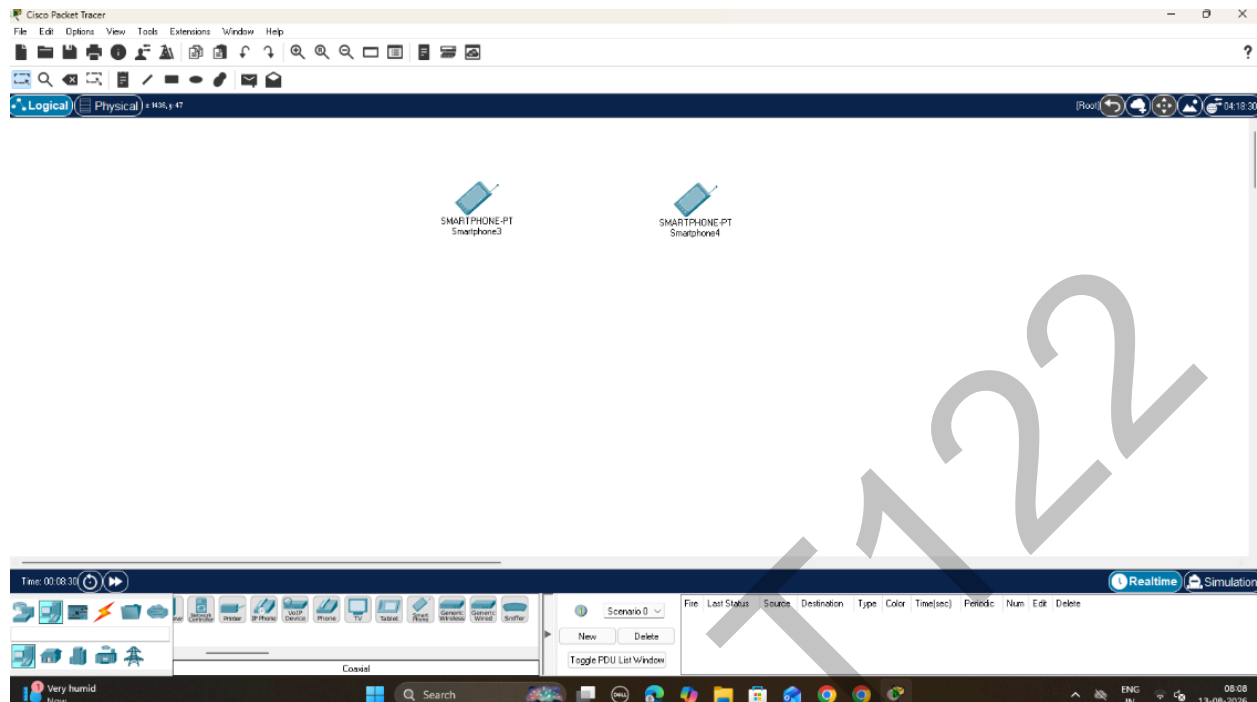


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Aim : Practical 6 – Cellular Network Simulation with Client-Server Communication

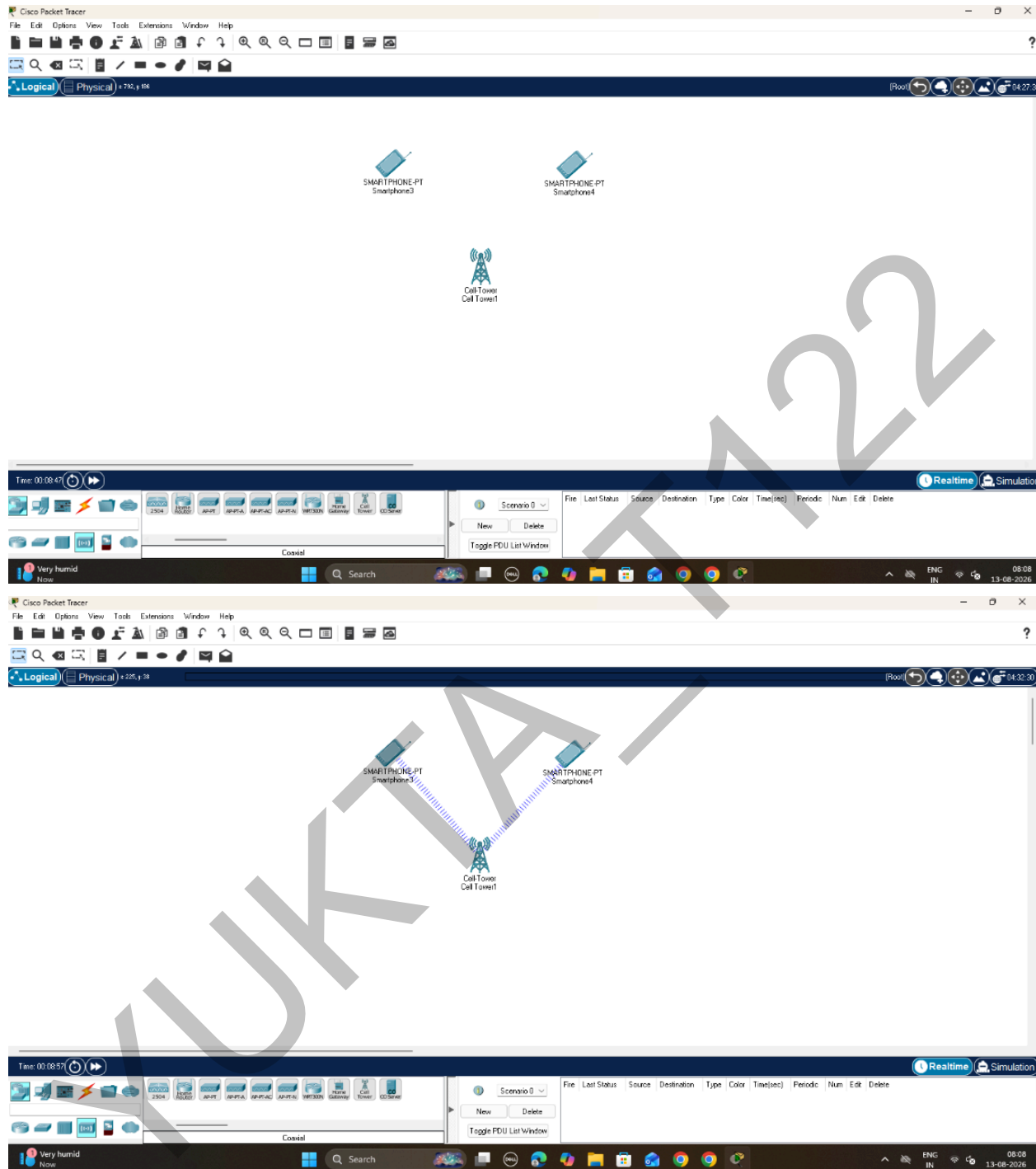


Steps

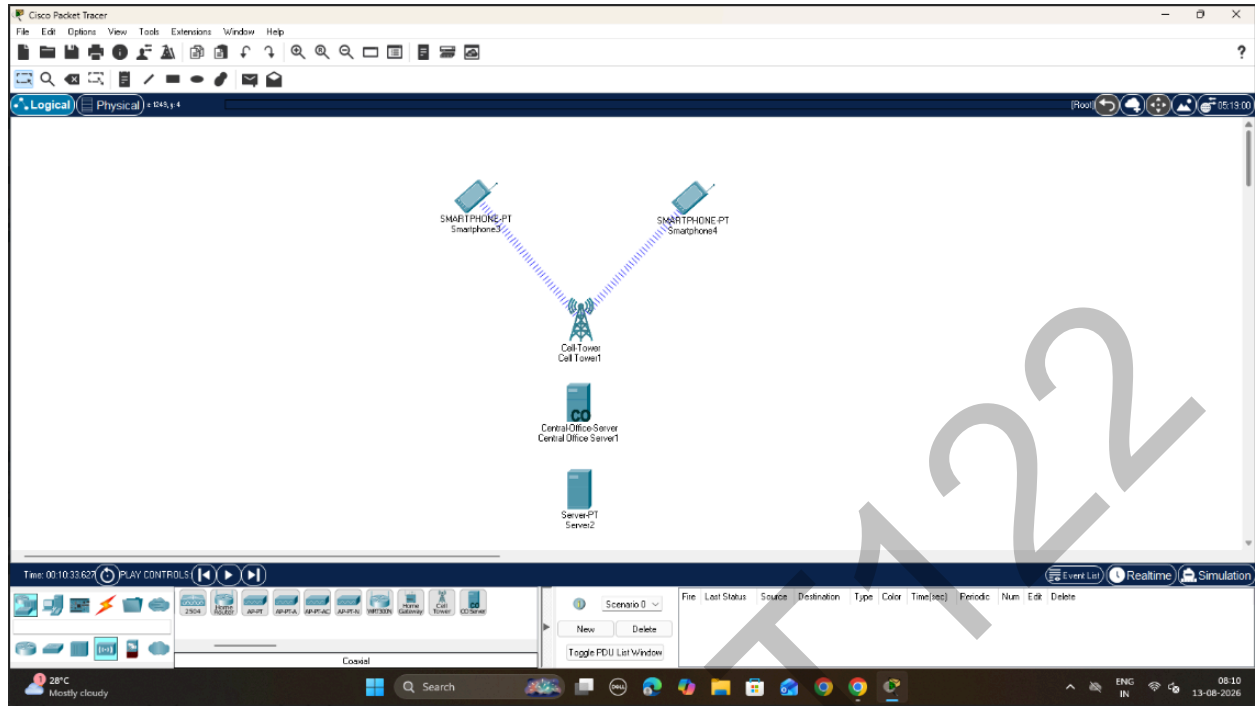
Part A – Create the Network Topology

1. Open **Cisco Packet Tracer**.
2. Create a new workspace.
3. From **End Devices**, add:
 - Server
 - PC/Laptop
 - Smartphone or other wireless/mobile devices.
4. From **Network Devices**, add a **Wireless Router/Access Point** or cellular network components available in your Packet Tracer version.
5. Connect the server to the network using an appropriate Ethernet cable.
6. Connect the wireless/mobile devices to the wireless network.

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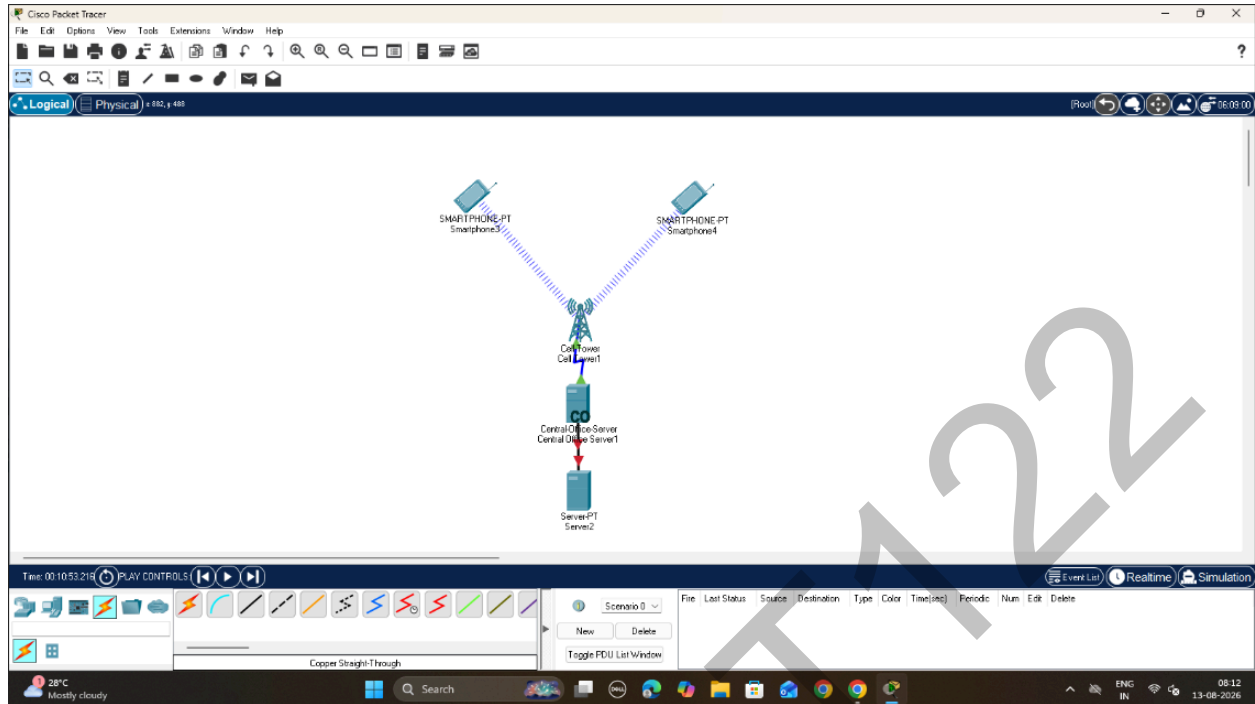
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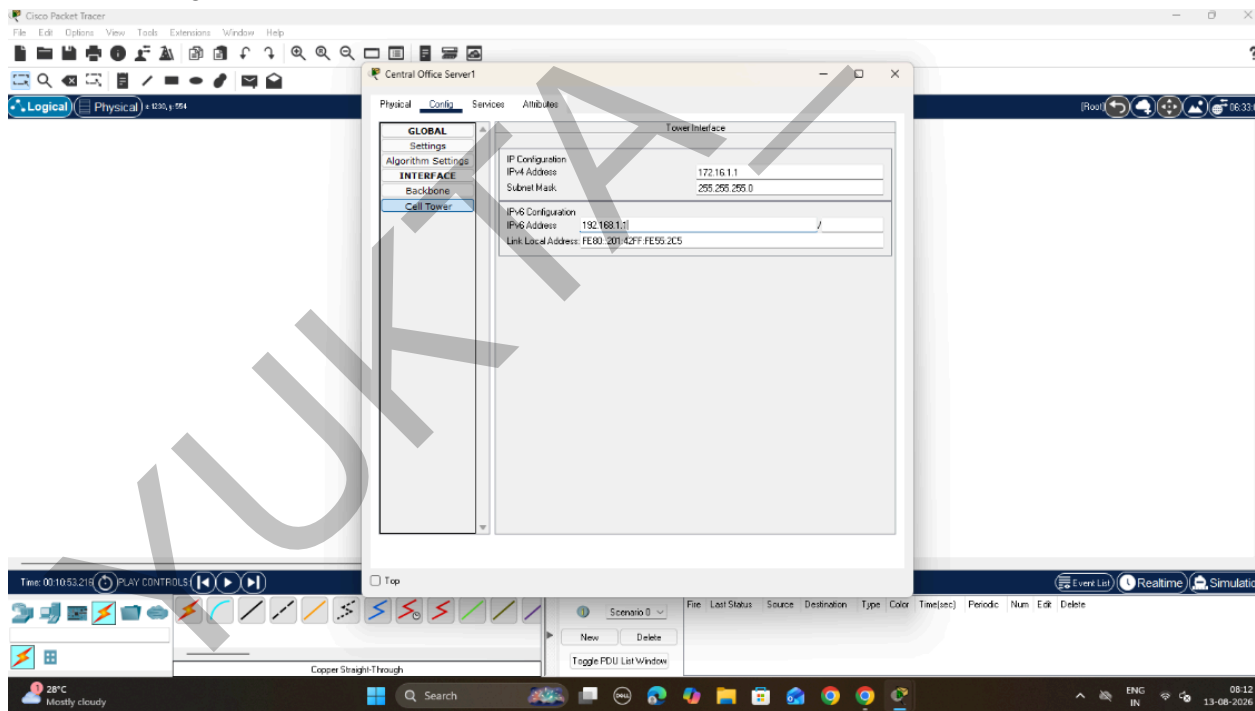
Part B – Configure the Server

7. Click the **Server**.
8. Go to **Desktop** → **IP Configuration**.
9. Assign an IP address, for example:
 - IP Address: 192.168.1.10
 - Subnet Mask: 255.255.255.0
 - Default Gateway: 192.168.1.1
10. Go to **Services**.
11. Enable the required service, such as **HTTP** for web-based client-server communication.
12. If required, configure **DHCP** on the server.
13. Create an appropriate DHCP pool, for example:
 - Default Gateway: 192.168.1.1
 - Start IP Address: 192.168.1.100
 - Subnet Mask: 255.255.255.0

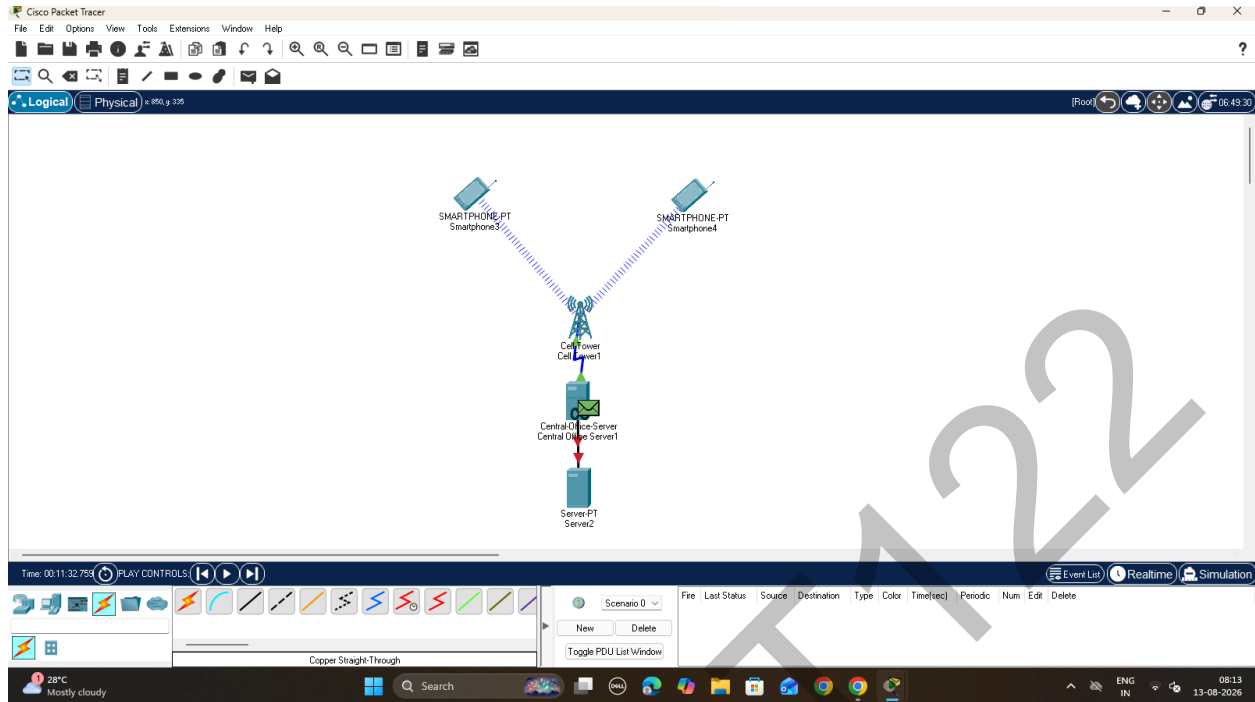
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Part C – Configure the Wireless/Cellular Network

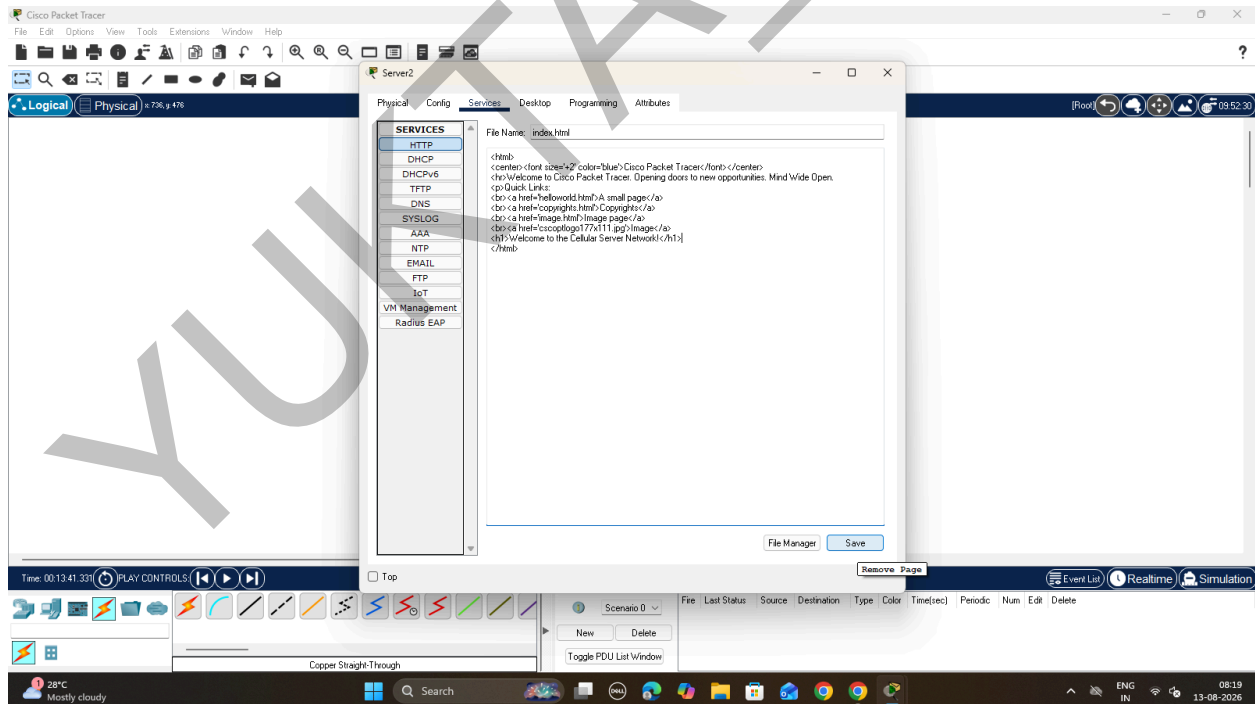


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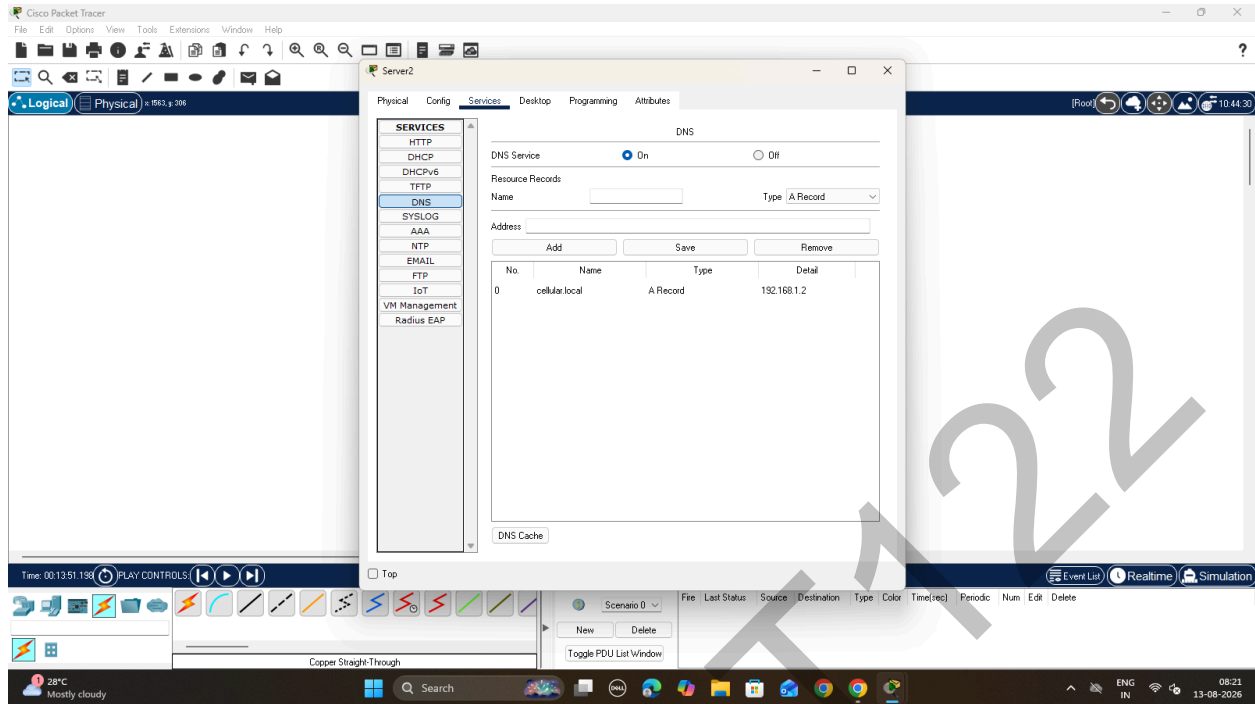


Part D – Configure the Client

Select the smartphone/laptop acting as the **client**.



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Part E – Test Connectivity

Open the client device.

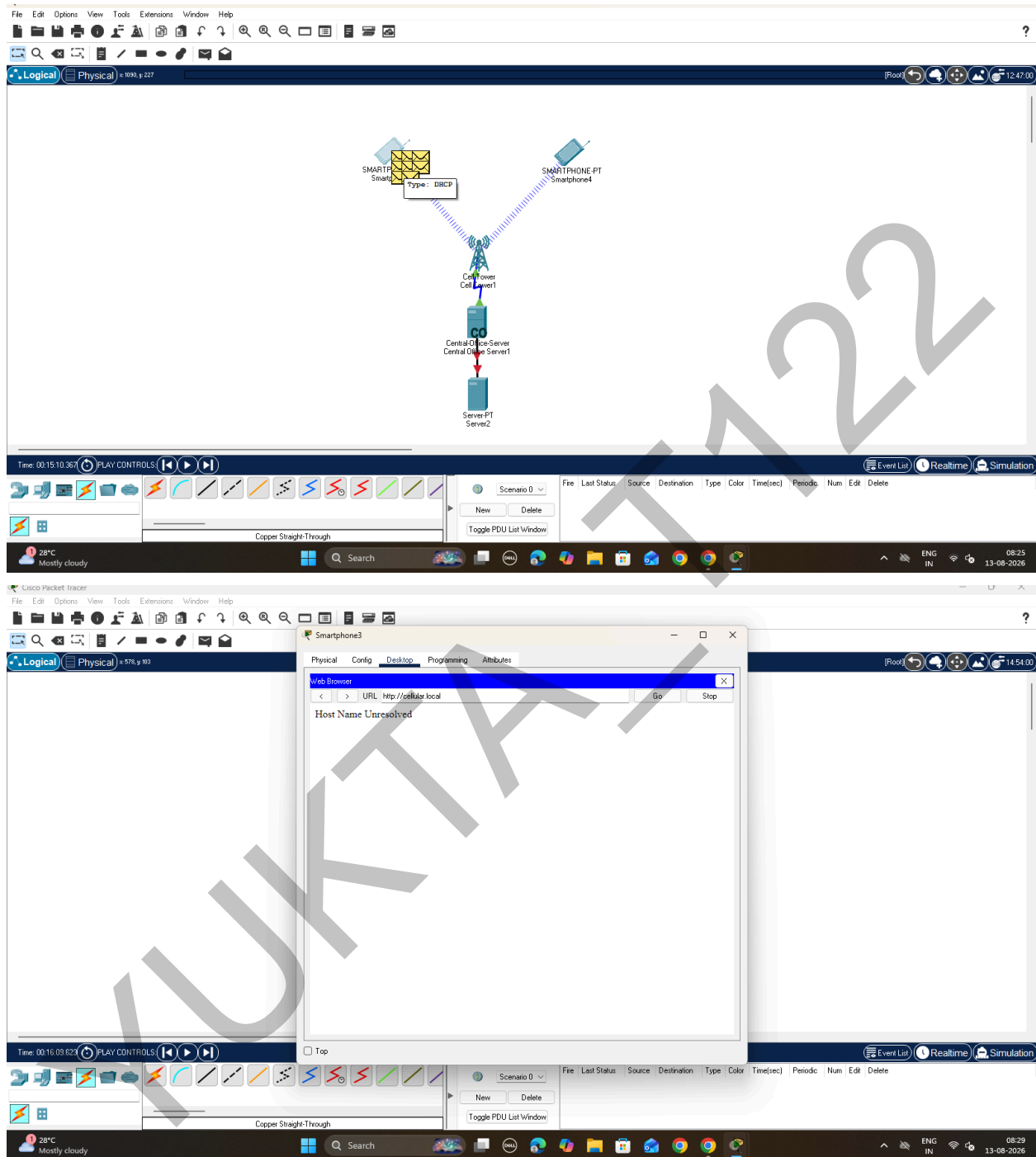
Go to **Command Prompt**.

Use the `ipconfig` command to check the client's IP configuration.

Test communication with the server using:

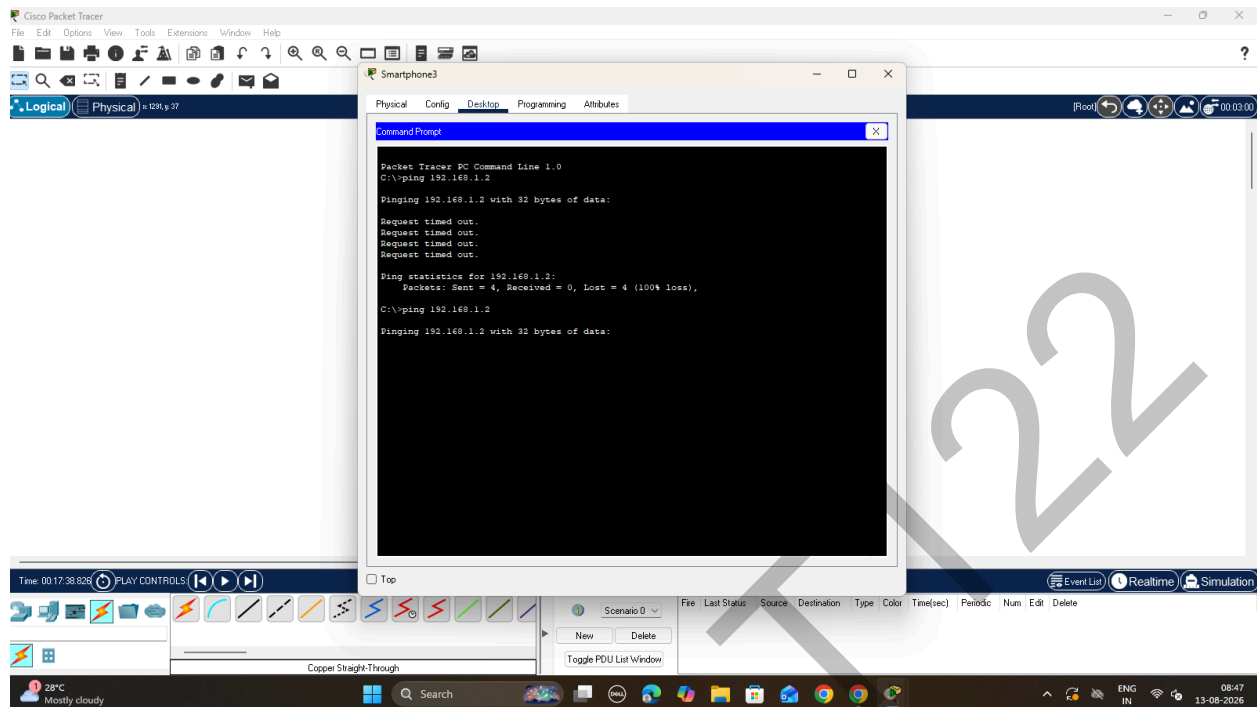
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Part F – Test Client-Server Communication



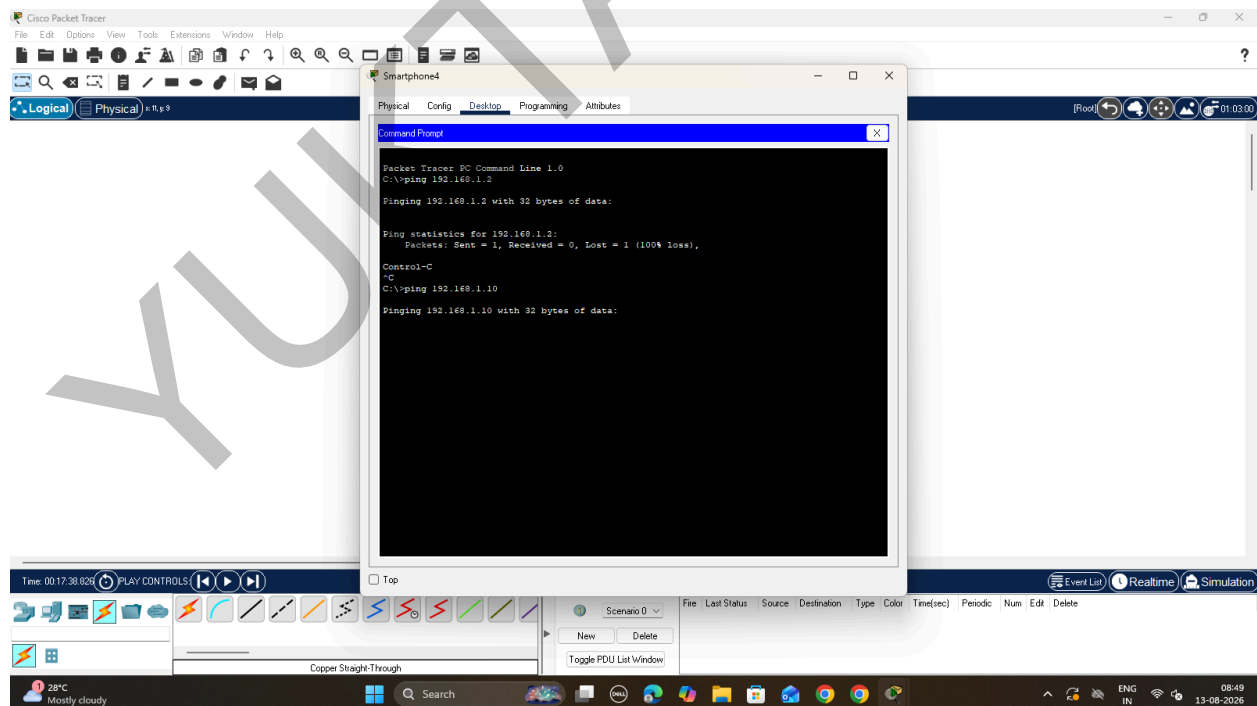
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Enter the server's IP address:



The server's HTTP page should appear.

This confirms that the client is communicating with the server through the simulated cellular/wireless network.



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Part G – Verify Using Simulation Mode

